Exact Synthesis of Two-Qubit Clifford+CS Circuits with a Near-Minimal Number of K and CS Gates

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We present a method for decomposing an arbitrary two-qubit Clifford+CS operator (equivalently, a 4×4 unitary with entries in $\mathbb{D}[i] = \mathbb{Z}[\frac{1}{2}, i]$) into a Clifford+CS circuit. Given a unitary U, let k be the least natural number such that all entries of $(1+i)^k U$ are Gaussian integers. Our method decomposes U, up to a global phase, into a circuit of length at most $20k + 9$, using at most $2k$ K gates (i.e., scaled H gates) and at most $2k + 1$ Controlled-S gates, without ancillae. Moreover, the K-count is at most 2 times the optimal, and the CS-count is at most 4 times the optimal. In experiments, this method always achieves the optimal CS-count, and yields on average a 19% K-count reduction compared to the best known result. Our approach is similar to that of Russell [35] for the Clifford+T gate set, but provides concrete bounds (e.g., $20k + 9$ vs. $O(k)$). Russell's method operates over the ring $\mathbb{Z}[\frac{1}{\sqrt{2}}, i]$, which is more complicated than and contains the ring $\mathbb{Z}[\frac{1}{2}, i]$ used here.

I. INTRODUCTION

In quantum computation, decomposing unitary operators into gates from a fixed universal set is a fundamental problem [33]. Like the Clifford+T gate set, the Clifford+Controlled-S gate set is also universal. We study the exact synthesis of 2-qubit Clifford+Controlled-S operators (the Controlled-S gate is abbreviated as CS). Previous work on two-qubit synthesis of Clifford+CS or Clifford+T operators has mainly focused on minimizing the use of the CS or T gate, since these gates are more expensive than Clifford operators in certain scenarios. However, work on T-gate optimization [1, 5, 9, 10, 21, 25] has suggested that the H gate is an "obstruction" to T-optimization methods. Quoting from [1]: "It has been shown that T-depth optimizations can be implemented efficiently for circuits consisting only of T and CNOT gates and that H gates impede the optimization significantly".

We propose a method that produces circuits with a near-minimal number of both K (i.e., scaled H) gates and CS gates, thereby alleviating this "obstruction". The synthesis problem of simultaneously minimizing the use of two different gates is also of mathematical interest. One reason is that previous work on single-qubit synthesis of Clifford+T operators achieved the minimum use of both Hadamard and T gates [14, 27, 31]; extending this to the multi-qubit case has been an open problem.

A. Background

Consider the following unitary operators:

$$i,\ K = \omega^7 H = \frac{1}{1+i}\begin{bmatrix}1 & 1\\ 1 & -1\end{bmatrix},\ S = \begin{bmatrix}1 & 0\\ 0 & i\end{bmatrix},\ \text{and}$$

$$CZ = \begin{bmatrix}1 & 0 & 0 & 0\\ 0 & 1 & 0 & 0\\ 0 & 0 & 1 & 0\\ 0 & 0 & 0 & -1\end{bmatrix}.$$

Here, i is the complex unit; $\omega = e^{\frac{i\pi}{4}}$ is the 8th root of unity; H is the Hadamard gate, and K is a scaled version of H; S is sometimes called the *phase gate*, and CZ is the controlled-Z gate. When closed under multiplication, identities, and tensor products, these operators generate the *Clifford group* (possibly up to scalars, depending on which scalars are included in the Clifford group; we only include the scalars ± 1 and $\pm i$ in our definition; hence, the above gates generate the Clifford group precisely).

Every operator U obtained in this way is of size $2^n \times 2^n$ for some natural number n, and as usual, we say that U is an operator on n qubits. It is well-known that the group of n-qubit Clifford operators is finite for any given n (see, e.g., [36]), and therefore not universal for quantum computing. We can obtain a universal gate set by adding the controlled phase gate

$$CS = \begin{bmatrix}1 & 0 & 0 & 0\\ 0 & 1 & 0 & 0\\ 0 & 0 & 1 & 0\\ 0 & 0 & 0 & i\end{bmatrix}.$$

The resulting operators are called the Clifford+CS operators. Note that $CZ = CS^2$. We write $\mathcal{CLI}$ for the group of 2-qubit Clifford operators; $\mathcal{CCS}$ for the 2-qubit Clifford+CS operators; and $\mathcal{CIR}$ for the set of 2-qubit Clifford+CS circuits (a circuit is a finite sequence of gates). Each circuit $D \in \mathcal{CIR}$ implements a unique operator in $\mathcal{CCS}$, denoted $[D]$. In other words, $[_]$ is the matrix semantics function from $\mathcal{CIR}$ to $\mathcal{CCS}$.

We use the following notations: we write $K_0 = K \otimes I, K_1 = I \otimes K$, and similarly for S_0 and S_1 where I is the 2×2 identity operator. We identify the scalar i with the 4×4-matrix $i\,(I \otimes I)$. In the notation for controlled gates, we use the convention that the target is the last index. For example, CX_{01} is a controlled X-gate

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with control qubit 0 and target qubit 1, also shortened to CX. CX_{10} is shortened to XC. The controlled S- and Z-gates are symmetric, in the sense that $CS_{jk} = CS_{kj}$ and $CZ_{jk} = CZ_{kj}$, and therefore the order of indices does not matter for them, so we just write CS and CZ. We also use the Z-gate, the X-gate, the controlled K-gate $CK = I \oplus K$, and the swap gate, denoted by Ex. They are related to the "primitive" gates introduced so far by $Z = S^2$, $X = KZKi$, $CX = K_1 CZ K_1 \cdot i$, $Ex = CX\, XC\, CX$, and

$$CK = CZ \cdot Z_0 S_0 Z_1 S_1 K_1 CS K_1 S_1 \cdot i. \tag{1}$$

Similarly to the CX gate, CK_{01} is shortened to CK and CK_{10} is shortened to KC. Note that by Equation 1, $CK \in \mathcal{CCS}$.

B. Related work

Operator synthesis can be done exactly or approximately. Here we only list a few of the most closely related works on exact synthesis.

1. Single-qubit and single-qudit

For single-qubit synthesis of Clifford+T operators, the works [14, 27, 31] achieved the minimum use of both the Hadamard and T gates. The works [16, 34] achieved the minimum use of the T gate for single-qutrit synthesis of Clifford+T operators. The work [23] considered single-qudit synthesis of Clifford+T operators. Forest et al. [12] generalized exact synthesis from Clifford+T to single-qubit Clifford-cyclotomic gate sets. Kliuchnikov and Yard [29] developed a general algebraic framework for exact synthesis over number rings, unifying several gate sets under one theory.

2. Two-qubit

Two works on two-qubit synthesis guarantee T-optimality or CS-optimality: 2-qubit Clifford+T synthesis [30] and 2-qubit Clifford+CS synthesis [17]. The first gave an algorithm that is near T-optimal (the worst case uses at most 10 times the minimum possible T gates). The latter gave an algorithm that is CS-optimal. Neither considered the H-count optimization, but we can deduce, for the latter work, that its H-count enjoys the same property as ours: the K-count is at most 2 times the optimal. Glaudell et al. [15] later gave a high-performance exact synthesis framework for two-qubit Clifford+T circuits that optimizes T-count exactly, using meet-in-the-middle with algebraic canonicalization and lookup tables. The above three works depend on the exceptional isomorphism between two Lie groups $SU(4) \cong \mathrm{spin}(6)$. Our work concerns the CS-optimality and H-optimality of 2-qubit Clifford+CS synthesis, but without using this exceptional isomorphism. Our work uses a similar "residue" analysis method to that of Russell [35], but over a simpler ring $\mathbb{Z}[\frac{1}{2}, i]$ contained in their ring $\mathbb{Z}[\frac{1}{\sqrt{2}}, i]$. Given a unitary U, Russell's method outputs a circuit of gate count $O(k)$, where k is the least natural number such that the entries of $(1+\omega)^k U$ are in the ring $\mathbb{Z}[\omega]$.

3. Multi-qubit and multi-qutrit

Multi-qubit synthesis of Clifford+T operators was proposed by Giles and Selinger [13]. Their algorithm's worst-case gate count is $O\left(3^{2^n} nk\right)$. Later work [26] improved the gate count to $O\left(4^n nk\right)$. Glaudell et al. later gave exact multi-qubit synthesis for several subclasses of Clifford+T operators [4], such as the class of operators generated by $\{X, CX, CCX, H\}$, known as the Toffoli+Hadamard operators. Amy et al. adapted the method of [26] to the Toffoli+Hadamard operators, achieving a similar improvement in gate count [3]. None of the works on multi-qubit synthesis introduced so far have considered T-count optimality. Kliuchnikov and Schönnenbeck [28] recently developed an optimal search algorithm for multi-qubit exact synthesis using A^* search. In particular, for two-qubit circuits, the runtime is proportional to the T-count. The works [2, 11] considered exact synthesis of multi-qubit Clifford+Cyclotomic operators. The works [19, 24] considered exact synthesis of multi-qutrit Clifford+Cyclotomic operators. The work [18] proved that qutrit metaplectic gates are a subset of Clifford+T. The work [32] proposed a heuristic algorithm that achieves V-gate optimality.

II. SOME ALGEBRA

A. Rings

As usual, we write $\mathbb{N} = \{0, 1, \ldots\}$ for the set of natural numbers. We also write $\mathbb{Z}$, $\mathbb{R}$, and $\mathbb{C}$ to denote the rings of integers, real numbers, and complex numbers, respectively. We write $z^\dagger$ for the complex conjugate of a complex number. Recall that in any ring, a *unit* is an invertible element.

Let $\mathbb{Z}[i] = \{a + bi \mid a, b \in \mathbb{Z}\}$ be the ring of Gaussian integers. It is a Euclidean domain[6], and therefore, division with remainder, greatest common divisors, divisibility, and the concept of a prime (i.e. a non-unit that cannot be written as the product of two non-units) all can be used in $\mathbb{Z}[i]$. Let $\gamma = 1 + i$. It is well-known that γ is a Gaussian prime [22]. Note that γ is a divisor of 2, and γ^2 and 2 divide each other. Using Euclidean division, it is further easy to check that

$$\mathbb{Z}[i]/(\gamma) = \{0, 1\}, \quad \mathbb{Z}[i]/(2) = \mathbb{Z}[i]/(\gamma^2) = \{0, 1, i, 1+i\},$$

$$\text{and} \quad \mathbb{Z}[i]/(\gamma^3) = \{\pm 1, \pm i, 0, 1+i, 1-i, 2\}.$$

Note that the above is a slight abuse of notation. Normally, $\mathbb{Z}[i]/(\gamma)$ denotes the quotient ring of $\mathbb{Z}[i]$ modulo the ideal (γ) whose elements are residues normally denoted by $x \bmod(\gamma)$ for $x \in \mathbb{Z}[i]$. Here we omit the mod (γ) for simplicity. Note that $\{0,1\}$ etc. are complete sets of residue class representatives and also the remainders from the Euclidean division of x by γ.

Let $\mathbb{D} = \mathbb{Z}[\frac{1}{2}] = \left\{\frac{a}{2^k} \mid a \in \mathbb{Z}, k \in \mathbb{N}\right\}$ be the ring of *dyadic fractions*, i.e., fractions whose denominator is a power of 2. Consider the ring $\mathbb{D}[i] = \{r + si \mid r, s \in \mathbb{D}\} = \mathbb{Z}[\frac{1}{2}, i]$. For every $t \in \mathbb{D}[i]$, there exists some natural number k such that $\gamma^k t \in \mathbb{Z}[i]$. This motivates the following definition.

Definition II.1. Let $t \in \mathbb{D}[i]$. A natural number $k \in \mathbb{N}$ is called a *denominator exponent* for t if $\gamma^k t \in \mathbb{Z}[i]$. The least such k is called the *least denominator exponent* of t, denoted by $\mathrm{lde}(t)$. Equivalently, the least denominator exponent of t is the smallest $k \in \mathbb{N}$ such that t can be written in the form $\frac{s}{\gamma^k}$, where $s \in \mathbb{Z}[i]$.

More generally, we say that k is a denominator exponent for a vector or matrix if it is a denominator exponent for all of its entries. The least denominator exponent for a vector or matrix is therefore the least k that is a denominator exponent for all of its entries.

Remark II.2. By enumerating all 2-qubit Clifford operators, we have that the lde of a 2-qubit Clifford operator is at most 2.

B. Residue rings

It is well-known that

$$\mathbb{Z}[i]/(\gamma^n) = \left\{\sum_{i=0}^{n-1} b_i\gamma^i \bmod (\gamma^n) \,|\, b_i \in \{0,1\}\right\}.$$

We give an elementary proof by induction on n. Note that a similar proof is used for the case $\mathbb{Z}[\sqrt{2}]/(\sqrt{2}^n)$ by Clement[8]. When $n = 1$, the claim obviously holds. Suppose this is true for n, i.e., for any $x \in \mathbb{Z}[i]$, x is congruent to some $\sum_{i=0}^{n-1} b_i\gamma^i \bmod (\gamma^n)$, that is

$$x = \sum_{i=0}^{n-1} b_i\gamma^i + k\gamma^n \text{ for some } k.$$

Since $k = b_n + k'\gamma$ for some $b_n \in \{0,1\}$, substituting k in and some rewrite gives

$$x = \sum_{i=0}^{n} b_i\gamma^i + k'\gamma^{n+1}, \text{ i.e., } x \equiv \sum_{i=0}^{n} b_i\gamma^i \bmod (\gamma^{n+1}).$$

For simplicity, we omit the mod (γ^n) in $\sum_{i=0}^{n-1} b_i\gamma^i \bmod (\gamma^n)$. Since $b_i \in \{0,1\}$, we then write binary string $(b_0b_1...)_2$ for the summation, and the parenthesis and subscript $_2$ can be omitted if we can infer that $b_0b_1...$ is a residue in binary format from the context.

Moreover, residue classes are distinct, i.e. if $b_0b_1...b_{n-1}$ and $b'_0b'_1...b'_{n-1}$ are equivalent, i.e., $\sum_{i=1}^{n-1} b_i\gamma^i - \sum_{i=1}^{n-1} b'_i\gamma^i = k\gamma^n$ for some k, then $b_i = b'_i$. Modding (γ) on both sides gives $b_0 = b'_0$. Then cancelling b_0 and b'_0, and dividing both sides by γ give

$$\sum_{i=1}^{n-1} b_i\gamma^{i-1} - \sum_{i=1}^{n-1} b'_i\gamma^{i-1} = k\gamma^{n-1}$$

Then modding (γ) on both sides again gives $b_1 = b'_1$. Repeating this process proves $b_i = b'_i$.

We write $\rho_n(x)$ for the canonical projection from $\mathbb{Z}[i]$ to n-digit residues. From the proof, we see $\rho_n(x)$ is "stable" w.r.t increase of n, i.e., the first n digits of $\rho_{n+1}(x)$ is $\rho_n(x)$. Note that ρ_2 is "stable" under taking complex conjugation, i.e., $\rho_2(x^\dagger) = \rho_2(x)$.

We sometimes write $x \equiv_{\gamma^n} y$ as an abbreviation for $x \equiv y \bmod \gamma^n$. Modulo γ^3, we have

x	$\rho_3(x)$	x	$\rho_3(x)$	x	$\rho_3(x)$	x	$\rho_3(x)$
1	100	-1	101	i	111	$-i$	110
0	000	$1+i$	010	$1-i$	011	2	001

The (mod γ^n) residue classes, as the quotient ring, is a commutative ring. In particular, modulo γ^2, the addition of the ring coincides with the bitwise exclusive or (denoted by $\oplus$) operation; $(10)_2$ is the ring identity; $(00)_2$ the zero of the ring; multiplication by $(01)_2$ is the right shift operation; by distributivity $11 \cdot b_0b_1 = 10 \cdot b_0b_1 + 01 \cdot b_0b_1 = b_0(b_1 \oplus b_0)$. Note that since $\rho_2(i) = 11$, if $\rho_1(x) = 1$, then either $\rho_2(x) = 10$ or $\rho_2(ix) = 10$. Similarly, modulo γ^3, the addition is

$$b_0b_1b_2 + b'_0b'_1b'_2 = (b_0 \oplus b'_0)(b_1 \oplus b'_1)(b_2 \oplus b'_2 \oplus (b_0 \cdot b'_0)),$$

where the $\cdot$ is the multiplication of the ring $\mathbb{Z}[i]/(\gamma)$, equivalently the "and" of two bits. The ring identity is 100, zero is 000. Multiplication by 010 is a right shift by 1 digit and multiplication by 001 is a right shift by 2 digits. The rest of the multiplication table can be obtained using distributivity. The additive inverse is

$$-(b_0b_1b_2) = b_0b_1(b_2 \oplus b_0).$$

By looking at the ρ_3 table, we see ρ_3 is not "stable" under taking complex conjugation, but instead, if $\rho_3(x) = abc$, then $\rho_3(x^\dagger) = ab\bar{c}$. Since $\rho_3(i) = 111$, if $\rho_1(x) = 1$, then there exists $k \in \{0,1,2,3\}$ such that $\rho_3(i^k x) = 100$.

Definition II.3. For $x \in \mathbb{Z}[i]$, if $\rho_1(x) = 0$, we say x is *even*, otherwise *odd*.

For example, $2 + 3i$ and $5 + 2i$ are odd, and $2 + 4i$ and $1 + 3i$ are even. In particular, in our context, by an even Gaussian integer we mean one that is divisible by γ, not necessarily by 2. However if $x \in \mathbb{Z}$, then x is

even (odd) as an integer if and only if x is even (odd) as a Gaussian integer. Therefore, our definition of parity extends the usual one on the integers. The following lemmas are straightforward.

Lemma II.4. *Let $\alpha = a + bi \in \mathbb{Z}[i]$. Then α is odd if and only if $\|\alpha\|^2 = a^2 + b^2$ is an odd integer.*

Lemma II.5. *If $\alpha \in \mathbb{Z}[i]$ is odd, then $\alpha \equiv \pm 1, \pm i \pmod{\gamma^3}$. In other words, $\alpha \equiv i^e \pmod{\gamma^3}$ for some $e \in \{0, 1, 2, 3\}$. Moreover, $\alpha \equiv i^e \pmod{\gamma^2}$ for some $e \in \{0, 1\}$.*

Lemma II.6. *If $\alpha, \beta \in \mathbb{Z}[i]$ satisfy $\alpha\gamma \equiv_{\gamma^{n+1}} \beta\gamma$, then $\alpha \equiv_{\gamma^n} \beta$.*

Let k be a denominator exponent (not necessarily the least) for $x \in \mathbb{D}[i]$, we define the residue function $\rho_n^k(x) = \rho_n(\gamma^k x)$. We call $\rho_n^k(x)$ the (n, k) residue of x. If k is a denominator exponent for a vector or matrix A, $\rho_n^k(A)$ is defined entry-wise.

Lemma II.7. *For $t \in \mathbb{D}[i]$, if $k = lde_\gamma(t) > 0$, then $\gamma^k t$ is odd.*

Lemma II.8. *For any $x \in \mathbb{D}[i]$ with denominator exponent k, $\rho_2^k(x^\dagger) = \rho_2^k(x)$ and $\rho_3^k(x^\dagger) = \rho_2^k(x) + 001$.*

C. Action on residues

Suppose $\rho_2(x) = b_0 b_1$, then $\rho_2(ix) = b_0(b_1 \oplus b_0)$. We say i acts on $b_0 b_1$ by sending it to $b_0(b_1 \oplus b_0)$. Given an n residue $b_0 b_1 ... b_{n-1}$, a right shift of it is an $n+1$ residue $0 b_0 b_1 ... b_{n-1}$. Given an $n+1$ residue $0 b_0 b_1 ... b_{n-1}$, a left shift is $b_0 b_1 ... b_n$. Note that we choose to only allow "well-defined" shifts in the sense no information is lost and no unknown is introduced. Note that the usual right shift is used previously when explaining the multiplication of $\mathbb{Z}[i]/(\gamma^2)$ and $\mathbb{Z}[i]/(\gamma^3)$. We use $LS(x)$ for the left shift of x and RS for the right shift.

Lemma II.9. *For $t \in \mathbb{D}[i]$, k a denominator exponent of t, we have*

$$\rho_{n+1}^{k+1}(t) = \rho_{n+1}^k(\gamma t) = RS(\rho_n^k(t)) \text{ and}$$

and if $\gamma^k t$ is even,

$$\rho_n^{k-1}(t) = \rho_n^k(t/\gamma) = LS(\rho_{n+1}^k(t)).$$

Given $x, y \in \mathbb{D}[i]$ with lde $k > 0$, we know $\rho_2^k(x) = 1b$ and $\rho_2^k(y) = 1b'$ for some $b, b' \in \{0, 1\}$. So $\rho_2^k(x + y) = 1b + 1b' = 0(b \oplus b')$. Then $\rho_1^{k-1}(x + y) = LS(\rho_2^k(x + y)) = b \oplus b'$. Similarly, $\rho_1^{k-1}(x - y) = LS(\rho_2^k(x - y)) = LS(1b - 1b') = LS((1 - 1)(b - b')) = b \oplus b'$ (subtraction in $\mathbb{Z}/(2)$ coincides with $\oplus$, in other words the additive inverse of b is itself). Note that if $b \oplus b' = 1$, then lde$(x \pm y)$ decreases by 1, and if $b \oplus b' = 0$, lde$(x \pm y)$ decreases by at least 2 if $k \geq 2$. Since dividing t by γ obviously increases lde(t) by 1, we have for $x, y \in \mathbb{D}[i]$ with lde $k > 0$ and $\rho_3^k(x) = 10b$ and $\rho_3^k(y) = 10b'$,

$$\rho_2^k(K \begin{bmatrix} x \\ y \end{bmatrix}) = \rho_2^k(\begin{bmatrix} (x+y)/\gamma \\ (x-y)/\gamma \end{bmatrix}) = LS(\rho_3^k(\begin{bmatrix} x+y \\ x-y \end{bmatrix})) = \begin{bmatrix} 0(b \oplus b' \oplus 1) \\ 0(b \oplus b') \end{bmatrix}.$$

So we have lde$(K \begin{bmatrix} x \\ y \end{bmatrix}) = k - 1$ and one entry of it has lde $k - 1$ and the other at most $k - 2$ if $k > 2$.

In general for $x, y \in \mathbb{D}[i]$ with lde $k > 0$ and $\rho_3^k(x) = 1b_0b_1$ and $\rho_3^k(y) = 1b'_0b'_1$,

$$\rho_2^k(K \begin{bmatrix} x \\ y \end{bmatrix}) = \rho_2^k(\begin{bmatrix} (x+y)/\gamma \\ (x-y)/\gamma \end{bmatrix}) = LS(\rho_3^k(\begin{bmatrix} x+y \\ x-y \end{bmatrix})) = \begin{bmatrix} (b_0 \oplus b'_0)(b_1 \oplus b'_1 \oplus 1) \\ (b_0 \oplus b'_0)(b_1 \oplus b'_1) \end{bmatrix}.$$

If $\rho_3^k(x) = 0b_0b_1$ and $\rho_3^k(y) = 0b'_0b'_1$,

$$\rho_2^k(K \begin{bmatrix} x \\ y \end{bmatrix}) = \begin{bmatrix} (b_0 \oplus b'_0)(b_1 \oplus b'_1) \\ (b_0 \oplus b'_0)(b_1 \oplus b'_1) \end{bmatrix}.$$

If $\rho_3^k(x) = 1b_0b_1$ and $\rho_3^k(y) = 0b'_0b'_1$,

$$\rho_3^{k+1}(K \begin{bmatrix} x \\ y \end{bmatrix}) = \begin{bmatrix} 1(b_0 \oplus b'_0)(b_1 \oplus b'_1) \\ 1(b_0 \oplus b'_0)(b_1 \oplus b'_1) \end{bmatrix}.$$

If $\rho_3^k(x) = 0b_0b_1$ and $\rho_3^k(y) = 1b'_0b'_1$,

$$\rho_3^{k+1}(K \begin{bmatrix} x \\ y \end{bmatrix}) = \begin{bmatrix} 1(b_0 \oplus b'_0)(b_1 \oplus b'_1) \\ 1(b_0 \oplus b'_0)(b_1 \oplus b'_1 \oplus 1) \end{bmatrix}.$$

III. SPECIAL UNITARIES

A. Permutation matrices

Let $a_0a_1a_2a_3$ be a permutation of $\{0, 1, 2, 3\}$. The permutation matrix $P_{a_0a_1a_2a_3}$ is the unitary matrix that maps e_i to e_{a_i}, where $\{e_i\}$ is the standard basis. All 24 permutation matrices are Clifford operators and can be implemented by circuits of length at most 3, see Table I.

B. Diagonal unitary matrices

The only diagonal unitary matrices with entries in $\mathbb{D}[i]$ (and $\mathbb{Z}[i]$) are of the form

$$\text{diag}(i^a, i^b, i^c, i^d), \text{ where } a, b, c, d \in \{0, 1, 2, 3\}.$$

P	circuit	P	circuit	P	circuit
0123	identity	1023	$X_1 \cdot CX$	2103	$X_0 \cdot XC$
1203	$Ex \cdot X_0 \cdot XC$	2013	$Ex \cdot X_1 \cdot CX$	0213	Ex
3210	$X_1 \cdot X_0$	2310	$X_0 \cdot CX$	2130	$Ex \cdot X_1 \cdot XC$
3120	$Ex \cdot X_1 \cdot X_0$	1320	$Ex \cdot X_0 \cdot CX$	1230	$X_1 \cdot XC$
3012	$XC \cdot X_1$	0312	$Ex \cdot XC$	0132	CX
3102	$Ex \cdot CX \cdot X_0$	1302	$Ex \cdot X_0$	1032	X_1
3021	$Ex \cdot XC \cdot X_1$	0321	XC	0231	$Ex \cdot CX$
3201	$CX \cdot X_0$	2301	X_0	2031	$Ex \cdot X_1$

TABLE I. Circuit implementation of 24 permutation matrices.

They can all be implemented by Clifford+CS circuits with length of at most 6 up to a global phase i^k, using at most 1 CS gate. This is because

$$Z_0^{b_0} Z_1^{b_1} S_0^{b_2} S_1^{b_3} CZ^{b_4} CS^{b_5} \cdot i^k, b_i \in \{0,1\}, k \in \{0,1,2,3\},$$

are all distinct and diagonal, and their number is 4^4, which equals the cardinality of the set of diagonal unitary matrices.

C. Generalized permutation unitary matrices

A generalized permutation unitary matrix with entries in $\mathbb{D}[i]$ is a matrix with the same nonzero pattern as a permutation matrix (there is exactly one nonzero entry in each row and each column), but the nonzero entry can be any number of unit length, i.e, powers of i. We will call such a matrix a generalized permutation. It is easy to see each 4×4 generalized permutation is product of a permutation matrix and a diagonal unitary matrix, hence can be implemented by a circuit of length at most 9 containing at most 1 CS gate and no K gate. Also, the group of generalized permutations is a semi-direct product of the group of permutations and the group of diagonal unitaries. In particular, for any permutation matrix P and diagonal matrix D, there exists D' such that

$$PD = D'P$$

and the CS-count (see Definition IV.2) of D and D' are the same.

IV. THE EXACT SYNTHESIS ALGORITHM

The method is similar to that of Russell [35] for the ring $\mathbb{Z}[\frac{1}{\sqrt{2}}, i]$, which is more complicated than and contains the ring $\mathbb{Z}[\frac{1}{2}, i]$ used here. Such residue analysis methods are widely used in exact synthesis, e.g., in [3, 13, 14, 17, 30]. Note that, compared to the ad hoc use of residues in each work, we have given in Section II a uniform way of dealing with residues of different rings using binary residues, which we conjecture works for any ξ-ring proposed in [28].

We start with the main theorem.

A. Main theorem

Theorem IV.1. *For any $A \in U(\mathbb{D}[i])$ with lde k, there exists a circuit C implementing A, using at most $2k$ K gates, at most $2k+1$ CS gates, and at most $9(2k+1)+2k$ gates up to a global phase.*

The proof, given later, is constructive, so it gives an algorithm for computing such circuit.

B. Optimality

Definition IV.2. Let D be a Clifford+CS circuit. Recall that $[D]$ is the unitary operator it implements. Define

- $\mathrm{rkc}(D)$ = the number of K gates that appear in D.
- $\mathrm{kc}(D) = \min\{\mathrm{rkc}(C) \mid C \in \mathcal{CIR} \text{ and } [C] = [D]\}$.
- $\mathrm{rcs}(D)$ = the number of CS gates that appear in D.
- $\mathrm{cs}(D) = \min\{\mathrm{rcs}(C) \mid C \in \mathcal{CIR} \text{ and } [C] = [D]\}$.
- $\mathrm{rlen}(D)$ = the number of gates that D contains.
- $\mathrm{len}(D) = \min\{\mathrm{rlen}(C) \mid C \in \mathcal{CIR} \text{ and } [C] = [D]\}$.
- by abuse of notation, $\mathrm{cs}, \mathrm{kc}, \mathrm{len}$ are also defined on operators that are implementable by circuits.

We say D is K-optimal, CS-optimal, and shortest if $\mathrm{rkc}(D) = \mathrm{kc}(D)$, $\mathrm{rcs}(D) = \mathrm{cs}(D)$, and $\mathrm{rlen}(D) = \mathrm{len}(D)$ respectively. We call $\mathrm{rkc}(D)$ and $\mathrm{rcs}(D)$ the (raw) K-count and (raw) CS-count of D respectively, and $\mathrm{kc}(D)$, $\mathrm{cs}(D)$ the optimal K-count and the optimal CS-count of D respectively.

Remark IV.3. Note that this convention differs from the one used in e.g. [20], where they would call, e.g., $\mathrm{cs}(D)$ the CS-count of D. In other words, their convention has no definition for the "raw" K-count (rkc) or CS-count (rcs), and the "count" they use is always the optimal one.

Any Clifford+CS circuit D is an alternating sequence of Clifford parts C_i and a CS gate

$$D = C_0 \cdot CS \cdot C_1 \cdot CS \cdot \ldots \cdot C_n. \tag{2}$$

By enumerating all 2-qubit Clifford circuits, we know C_i is implementable by at most 2 K gates and $\mathrm{lde}(C_i) \leq 2$. By subadditivity of lde, we have $\mathrm{lde}(D) \leq \mathrm{lde}(C_0) +$

$\mathrm{lde}(CS)+\mathrm{lde}(C_1)\ldots\mathrm{lde}(C_n) \leq 2(n+1)$, that is, $\mathrm{lde}(D) \leq (2(\mathrm{rcs}(D)+1)$ which implies

$$\mathrm{lde}(D) \leq 2(\mathrm{cs}(D)+1).$$

Suppose 2 is CS-optimal, i.e., $\mathrm{cs}(D) = n$, we have $\mathrm{rkc}(D) \leq 2(\mathrm{cs}(D)+1)$ which implies

$$\mathrm{kc}(D) \leq 2(\mathrm{cs}(D)+1).$$

Any Clifford+CS circuit D is also an alternating sequence of generalized permutation parts P_i and a K_1 gate (by replacing all occurrences of K_0 by $Ex \cdot K_1 \cdot Ex$):

$$D = P_0 \cdot K_1 \cdot P_1 \cdot K_1 \cdot \ldots \cdot P_n. \tag{3}$$

Each generalized permutation is implementable by at most 1 CS gate and uses no K gate. So we have $\mathrm{cs}(D) \leq \mathrm{rkc}(D)+1$, which implies

$$\mathrm{cs}(D) \leq \mathrm{kc}(D)+1.$$

We also have, by subadditivity of lde, that $\mathrm{lde}(D) \leq \mathrm{rkc}(D)$, which implies

$$\mathrm{lde}(D) \leq \mathrm{kc}(D).$$

Theorem IV.1 outputs D' such that $\mathrm{rkc}(D') \leq 2\mathrm{lde}(D)$ and $\mathrm{rcs}(D') \leq 2\mathrm{lde}(D)+1$, which, together with the previous inequalities, implies

$$\mathrm{lde}(D) \leq \mathrm{kc}(D) \leq 2\mathrm{lde}(D) \tag{4}$$

$$\mathrm{kc}(D)/2-1 \leq \mathrm{cs}(D) \leq 2\mathrm{lde}(D)+1 \tag{5}$$

From above inequalities, our synthesis algorithm is at worst 2 times the optimal K-count, and at worst 4 times the optimal CS-count (experiments show it always achieves the optimal CS-count and a better K-count than the best known; see Section V). Note that Inequality 5 is similar to Proposition 6 of [17], but slightly tighter.

1. *K optimality of Glaudell et al.'s algorithm*

The work of Glaudell et al. [17] did not consider the K-count at all, but we can derive from their CS optimality that for a unitary A,

$$\mathrm{cs}(A) \leq \mathrm{kc}(A) \leq 2\,\mathrm{cs}(A)+2.$$

This is because the output of their algorithm is of the form Equation 2 with the CS-count being optimal and the Clifford parts in between CS gates having K-count 1 or 2. Take a C_i that lies between two CS gates and call it E. First, $\mathrm{kc}(E) \neq 0$, because otherwise E is a generalized permutation which contains no CS gate (since it is Clifford). So the two CS gates in $CS \cdot E \cdot CS$ would cancel, decreasing the CS-count, which contradicts the optimality. The cancellation can be shown by direct calculation since the generalized permutations are finite, or by using the semi-direct product structure that the permutations and diagonals together form. We can show $\mathrm{kc}(E) \leq 2$ by enumerating all two-qubit Cliffords.

So Glaudell et al.'s algorithm enjoys the same property as ours: the K-count is at most 2 times worse than optimal.

Remark IV.4. Note that experiments show that in most cases $\mathrm{lde}(A) < \mathrm{cs}(A)$ for randomly generated A. So our bound $\mathrm{kc}(A) \leq 2\mathrm{lde}(A)$ is strictly better in practice.

C. Complexity

Since we need only finitely many operations (independent of the lde; proved later) to decrease the lde by 1, and the unitaries with lde 0 are generalized permutations, the time complexity is $O(\mathrm{lde}(U))$ for any input $U \in \mathcal{CCS}$.

D. Lemmas and proofs

We start with a number of lemmas. Since $\mathbb{Z}[i]$ and $\mathbb{D}[i]$ are subrings of $\mathbb{C}$, the usual properties of complex numbers, complex vectors, and complex matrices, such as complex conjugation (denoted by $_^\dagger$), inner product, norm, transpostion (denoted by $_^T$), and conjugate transposition (also denoted by $_^\dagger$) are naturally inherited. For example, for a vector $v \in \mathbb{D}[i]^n$, the *norm* of v is $\|v\| = \sqrt{v^\dagger v}$. Note that $\|v\|^2 = v^\dagger v \in \mathbb{D}$, since $v^\dagger v \in \mathbb{D}[i] \cap \mathbb{R}$. Similarly if $v \in \mathbb{Z}[i]$, then $v^\dagger v \in \mathbb{Z}$. A *unit vector* is a vector of norm 1.

If v is a vector, the notation v_j refers to its jth entry; entries are numbered starting from $j = 0$.

Lemma IV.5. *For any $A \in U(\mathbb{D}[i])$ with lde k, there exists permutation matrix L and R such that, $\rho_1^k(LAR)$ or $\rho_1^k(LAR)^T$ equals one of the following*

$$(i)\begin{bmatrix}1&0&0&0\\0&1&0&0\\0&0&1&0\\0&0&0&1\end{bmatrix},\quad (ii)\begin{bmatrix}1&1&0&0\\1&1&0&0\\0&0&0&0\\0&0&0&0\end{bmatrix},\quad (iii)\begin{bmatrix}1&1&0&0\\1&1&0&0\\0&0&1&1\\0&0&1&1\end{bmatrix},$$

$$(iv)\begin{bmatrix}1&1&1&1\\1&1&1&1\\0&0&0&0\\0&0&0&0\end{bmatrix},\quad (v)\begin{bmatrix}1&1&0&0\\1&1&0&0\\1&1&1&1\\1&1&1&1\end{bmatrix},\quad (vi)\begin{bmatrix}1&1&1&1\\1&1&1&1\\1&1&1&1\\1&1&1&1\end{bmatrix}.$$

We also say A has such pattern and write $\mathrm{pat}(A)$ *for the pattern of A. Moreover, A has pattern (i) if and only if $\mathrm{lde}(A) = 0$, i.e., when A is a generalized permutation. A has pattern (iv), (v), and (vi) only if $\mathrm{lde}(A) > 1$.*

Before proving this lemma, we prove several useful lemmas.

Lemma IV.6. *Let v be a unit vector in $\mathbb{D}[i]^4$ with lde $k > 0$. There exists a generalized unitary permutation matrix P such that $\rho_1^k(Pv) = (1,1,0,0)^T$ or $(1,1,1,1)^T$, $\rho_2^k(Pv) = (10,10,01,00)^T$ or $(10,10,10,10)^T$ provided $k > 1$, and $\rho_2^1(Pv) = (10,10,00,00)^T$ provided $k = 1$.*

Proof. Let $w = \gamma^k v \in \mathbb{Z}[i]^4$. We first show w has an even number of odd entries. We have

$$\sum_{j=0}^{3} \|w_j\|^2 = \|w\|^2 = |\gamma|^{2k}\|v\|^2 = (\gamma^\dagger\gamma)^k \equiv 0 \,(\mathrm{mod}\ \gamma).$$

Therefore, there are an even number of $j \in \{0, ..., 3\}$ such that $\|w_j\|^2$ is odd. It follows by Lemma II.4 that an even number of w_j are odd.

Then we can find a permutation matrix P such that $\rho_2^k(Pv) = (1a, 1b, 0c, 0d)^T$ or $(1a, 1b, 1c, 1d)^T$. For the latter, by multiplying a diagonal matrix, i.e., multiplying i to the required entries, we can make the second digit all zero, i.e., $(10, 10, 10, 10)^T$.

For the former, by multiplying a diagonal matrix, we can make the second digit of the odd entries zero, i.e., $(10, 10, 0c, 0d)^T$. Now applying CK (equivalently applying K to the pair of odd entries) to it, we get $(00, 01, 0c, 0d)^T$ or $(01, 00, 0c, 0d)^T$. Note that multiplication of CK and a unit vector is still a unit vector. So $(01, 00, 0c, 0d)^T$ or $(00, 01, 0c, 0d)^T$ is a residue of a unit vector with lde $k-1$, then its $(1, k-1)$ residue must of the form $(1, 1, 0, 0)^T$ up to a permutation if $k > 1$, that is $c = 1$ and $d = 0$.

If $k = 1$, the only unit vectors are $\frac{1}{\gamma}(i^l, i^m, 0, 0)^T$ for $l, m \in \{0, 1, 2, 3\}$ up to a permutation. □

Lemma IV.7. *Let u, v be two unit vectors in $\mathbb{D}[i]^4$ with lde $k > 0$ such that $u \perp v$, i.e., $u^\dagger v = 0$. Then there exists a generalized unitary permutation matrix Q such that $\rho_1^k(Q[u|v])$ equals*

$$\begin{bmatrix}1&1\\1&1\\0&0\\0&0\end{bmatrix}, \begin{bmatrix}1&0\\1&0\\0&1\\0&1\end{bmatrix}, \begin{bmatrix}1&1\\1&1\\1&0\\1&0\end{bmatrix}, \begin{bmatrix}1&1\\1&1\\0&1\\0&1\end{bmatrix}, \text{ or } \begin{bmatrix}1&1\\1&1\\1&1\\1&1\end{bmatrix}.$$

Proof. By Lemma IV.6, there can be only even number of 1s in each column. The only missing case is

$$\begin{bmatrix}1&0\\1&1\\0&1\\0&0\end{bmatrix}.$$

Then $\rho_1^k(u^\dagger v) = \sum_{j=0}^{3} \rho_1^k(\bar{u}_i v_i) = 1$ contradicts that $\rho_1^k(u^\dagger v) = \rho_1^k(0) = 0$. In other words, the number of indices such that at which the $(1, k)$ residue of u, v is 1 must be even. □

Proof of Lemma IV.5. Since the rows and columns of a unitary matrix are unit vectors and all row (column) vectors are perpendicular to each other. Lemma IV.6 and IV.7 forces the pattern to be (ii) - (vi) if $k > 0$. If $k = 0$, the only unitaries with lde 0 are the generalized permutations which has pattern (i). If $k = 1$, the only unit vectors are $\frac{1}{\gamma}(i^l, i^m, 0, 0)^T$ up to a permutation. So residue of such vectors cannot have four 1s. So Case (iv), (v), and (vi) only happens if the lde > 1. □

Lemma IV.8. *For any $A \in \mathcal{CCS}$ with lde $k > 0$, there exist generalized permutations L, R, L', R' such that one of the following holds:*

- *$lde(K_1LAR) \le k-1$;*
- *or $lde(LARK_1) \le k-1$;*
- *or $lde(L'K_1LARR'K_1) \le k-1$;*
- *or $lde(K_1L'K_1LARR') \le k-1$;*
- *or $lde(L'CKLAR) \le k-1$ (only happens when $lde(A) = 1$).*

Note that each $\le$ above is actually $=$, but we don't need this. The proof proceeds by a case distinction over $\rho_1^k(A)$. Note that requiring $k > 0$ eliminates Case i. So start with Case ii.

1. Case ii

If $k > 1$, by Lemma IV.6, up to multiplying generalized permutations on the left and right, pat(A) refines to

$$\begin{bmatrix}10&10&01&00\\10&1a&0b&0\bar{b}\\01&0c&0d&0e\\00&0\bar{c}&0f&0g\end{bmatrix},$$

where $\bar{b} = b \oplus 1$ and b, c, d, e, f, g are indeterminates (later we use indeterminates in a residue matrix without declaring them). We claim $a = 0, b = c = 1$. Taking inner product of the first two column gives

$$10 + 1a + 00 + 00 = 00 \Rightarrow a = 0.$$

Taking inner product of the second and third row gives $c = 1$; similarly, we have $b = 1$. Now pat(A) refines to

$$\begin{bmatrix}10&10&01&00\\10&10&01&00\\01&01&0d&0e\\00&00&0f&0g\end{bmatrix}.$$

Moreover, since $k > 1$ and the third and fourth row (column) is of lde $k-1 > 0$, so $d = f$, $d = e$, $e = g$ by Lemma IV.6. So $d = e = f = g$. Now pat(A) refines to

$$\begin{bmatrix}10&10&01&00\\10&10&01&00\\01&01&0d&0d\\00&00&0d&0d\end{bmatrix}.$$

Applying K_1S_1 on the left gives Case iv.

If $k = 1$, up to multiplying generalized permutations on the left and right, pat(A) refines to

$$\begin{bmatrix} 10 & 10 & 00 & 00 \\ 10 & 1a & 00 & 00 \\ 00 & 00 & 00 & 00 \\ 00 & 00 & 00 & 00 \end{bmatrix},$$

As before, we have $a = 0$. Apply $X_0 \cdot CK \cdot X_0$ on the left decreases the lde.

Remark IV.9. Note that the two generalized permutations we use to refine the pattern can be chosen to be CS-free. This is because the (2,k)-residue of the two rows (columns) of even entries are stable under multiplication of i and we can introduce "dummy" i at a proper place to make a generalized permutation CS-free.

2. *Case iii*

If $k > 1$, by Lemma IV.6, up to multiplying generalized permutations on the left and right, pat(A) refines to

$$\begin{bmatrix} 10 & 10 & 01 & 00 \\ 10 & 1a & 0b & 0\bar{b} \\ 01 & 0c & 10 & 10 \\ 00 & 0\bar{c} & 10 & 1d \end{bmatrix},$$

Taking inner product of the first two column gives $a = 0$. Taking inner product of the second and third row gives $c = 0$; similarly, we have $b = 0$. Now pat(A) refines to

$$\begin{bmatrix} 10 & 10 & 01 & 00 \\ 10 & 10 & 00 & 01 \\ 01 & 00 & 10 & 10 \\ 00 & 01 & 10 & 10 \end{bmatrix}.$$

Applying K_1S_1 on the left brings us to Case vi.

If $k = 1$, by Lemma IV.6, up to multiplying generalized permutations on the left and right, pat(A) refines to

$$\begin{bmatrix} 10 & 10 & 00 & 00 \\ 10 & 1a & 00 & 00 \\ 00 & 00 & 10 & 10 \\ 00 & 00 & 10 & 1d \end{bmatrix}.$$

As before, we have $a = 0, d = 0$. Applying K_1 on the left decreases the lde.

3. *Case iv*

This case only happens for $k > 1$. By Lemma IV.6, up to multiplying generalized permutations on the left and right, pat(A) refines to

$$\begin{bmatrix} 10 & 10 & 10 & 10 \\ 10 & 1a & 1b & 1c \\ 01 & 0d & 0e & 0f \\ 00 & 0\bar{d} & 0\bar{e} & 0\bar{f} \end{bmatrix}.$$

Taking inner product of the first two columns $a = 0$; similarly we have $b = c = 0$. Since the third row is of lde $k - 1 > 0$, either $d = e = f = 1$ or one of d, e, f is 1 and the rest are 0. For the latter, up to a permutation of columns, we may assume $d = 1, e = f = 0$. Then, for both cases, right multiplication of K_1 decreases the lde.

Remark IV.10. As in Remark IV.9, the left generalized permutation can be chosen to be CS-free.

4. *Case v*

This case only happens for $k > 1$. By Lemma IV.6, up to multiplying generalized permutations on the left and right, pat(A) refines to

$$\begin{bmatrix} 10 & 10 & 01 & 00 \\ 10 & 1a & 00 & 01 \\ 10 & 1b & 10 & 1f \\ 10 & 1c & 1e & 10 \end{bmatrix},$$

We claim $a = 0, b = c = e = 1$. Taking inner product of the first two rows gives $a = 0$. Taking inner product of the first and third row gives $b = 1$; similarly $c = 1$. Taking inner product of the second and third column gives $e = 1$; similarly $f = 1$. Now the pattern is

$$\begin{bmatrix} 10 & 10 & 01 & 00 \\ 10 & 10 & 00 & 01 \\ 10 & 11 & 10 & 11 \\ 10 & 11 & 11 & 10 \end{bmatrix},$$

Applying K_1 on the left brings us to Case iv.

5. *Case vi*

This case only happens for $k > 1$. By Lemma IV.6, up to multiplying generalized permutations on the left and right, pat(A) refines to

$$\begin{bmatrix} 10 & 10 & 10 & 10 \\ 10 & 1a & 1d & 1g \\ 10 & 1b & 1e & 1h \\ 10 & 1c & 1f & 1j \end{bmatrix},$$

Taking inner product of first two columns gives that a, b, c cannot be all 1. Up to a permutation, we may assume $a = 0$. Then $b = c$; similarly $d = g$.

If $b = c = 1$, and $d = g = 1$, pat(A) is

$$\begin{bmatrix} 10 & 10 & 10 & 10 \\ 10 & 10 & 11 & 11 \\ 10 & 11 & 1e & 1h \\ 10 & 11 & 1f & 1j \end{bmatrix},$$

Taking inner product of the first and third column gives $f = \bar{e}$; similarly, $h = \bar{e}$ and $j = \bar{f}$. Now pat(A) is

$$\begin{bmatrix} 10 & 10 & 10 & 10 \\ 10 & 10 & 11 & 11 \\ 10 & 11 & 1e & 1\bar{e} \\ 10 & 11 & 1\bar{e} & 1e \end{bmatrix}.$$

We prove this pattern cannot happen by looking at its (3,k)-residue. Up to multiplication of powers of -1 on the proper rows and columns, the (3,k)-residue refines to

$$\begin{bmatrix} 101 & 101 & 101 & 101 \\ 101 & 10x & 11u & 11p \\ 101 & 11y & 1ev & 1\bar{e}q \\ 101 & 11z & 1\bar{e}w & 1er \end{bmatrix}.$$

Taking inner product of the first column with the second and third column, we have

$$101 + 10x + 11y + 11z = 000 \implies x \oplus y \oplus z = 1,$$
$$101 + 11u + 1ev + 1\bar{e}w = 000 \implies u \oplus v \oplus w = 1.$$

Taking inner product of the second and third column gives

$$101 + 11(u \oplus \bar{x}) + (1ev + 01e + 00\bar{y}) + (1\bar{e}w + 01\bar{e} + 00\bar{z})$$
$$= 000 \implies x \oplus y \oplus z \oplus u \oplus v \oplus w = 1, \text{ a contradiction.}$$

If $b = c = 0$, taking inner product of the first and third row gives $e = h$, similarly, $f = j$. So pat(A) refines to

$$\begin{bmatrix} 10 & 10 & 10 & 10 \\ 10 & 10 & 1d & 1d \\ 10 & 10 & 1e & 1e \\ 10 & 10 & 1f & 1f \end{bmatrix},$$

Taking inner product of the first and third column gives $d \oplus e \oplus f = 0$, then at least one of d, e, f is 0. Up to a permutation of rows, we may assume $d = 0$. Similar analysis applies if $d = g = 0$. So, for either case, up to permutations of rows and columns pat(A) refines to

$$\begin{bmatrix} 10 & 10 & 10 & 10 \\ 10 & 10 & 10 & 10 \\ 10 & 10 & 1e & 1e \\ 10 & 10 & 1f & 1f \end{bmatrix},$$

Taking inner product of the first and third column implies $e = f$. So left multiplication of K_1 reduces the lde.

Thus we have exhausted all cases. The transitions are summarized in Figure 1.

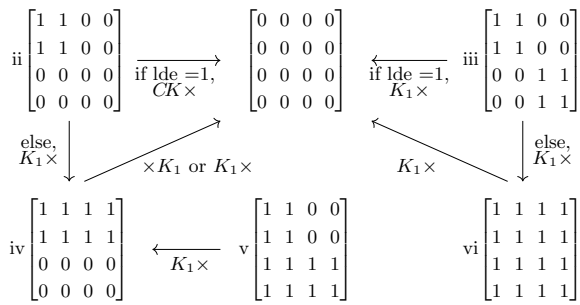

FIG. 1. Transitions between cases. The $K_1\times$ ($\times K_1$) annotation on an arrow means that if A has the pattern at the tail, then K_1LAR ($LARK_1$) has the pattern at the head for appropriate chosen generalized permutations L and R. Similarly for other annotated arrows. Note that CK decomposes into a circuit with two K gates as in Equation 1.

Proof of Theorem IV.1. Let $k = \text{lde}(A)$. By Lemma IV.8, by possibly merging adjacent generalized permutations, there exist generalized permutations $L_1, L_2, ..., L_p, R_1, R_2, ..., R_q$ such that

$$\text{lde}(L_pK_1...L_2K_1L_1AR_1K_1R_2K_1..., R_q) = 0.$$

That is, there exists a generalized permutation P such that

$$L_pK_1...L_2K_1L_1AR_1K_1R_2K_1..., R_q = P.$$

Note that the rkc of the left-hand side is $\leq 2k$. Then we have

$$A = L_1^{-1}iK_1L_2^{-1}iK_1...L_p^{-1}PR_q^{-1}iK_1...R_2^{-1}iK_1R_1^{-1}.$$

Note that the rkc of the right-hand side is at most $2k$. After merging adjacent generalized permutations, the number of generalized permutations is at most $2k + 1$. Each generalized permutation is implementable by at most 9 gates and at most 1 CS gate, up to a global phase. So we have a circuit that implements A, using at most $2k$ K gates, at most $2k+1$ CS gates, and at most $9(2k+1)+2k$ gates up to a global phase. □

V. EXPERIMENT

We implemented our algorithm in Haskell and compared it with Glaudell et al.'s CS-optimal synthesis algorithm [17]. Unitaries of a given lde range are randomly generated and fed to both algorithms, and the output circuits are compared on the metrics rcs and rkc. The output of their algorithm is of the form Equation 2 with the CS-count being optimal. We added one step to their algorithm to keep the K-count as small as possible: for each C_i, we replace it with an equivalent circuit of minimum K-count (which may be 0, 1, or 2). We randomly generated 1,346 two-qubit Clifford+CS operators with lde ranging from 2 to 58; data are available in VI. Figure 2 shows the distribution of lde values. Figure 3 shows

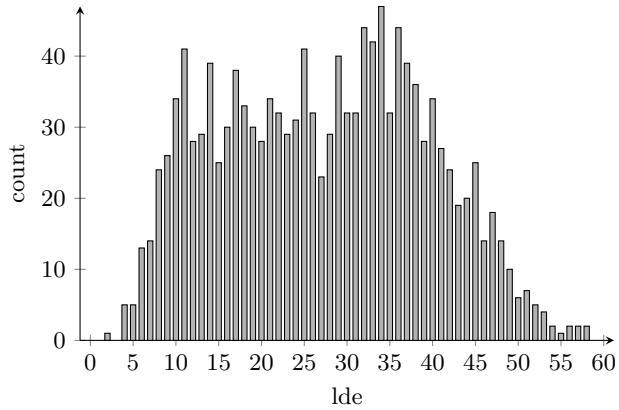

FIG. 2. Distribution of the least denominator exponent (lde) across the 1,346 test operators. The lde ranges from 2 to 58, with a mean of 27 and a median of 28.

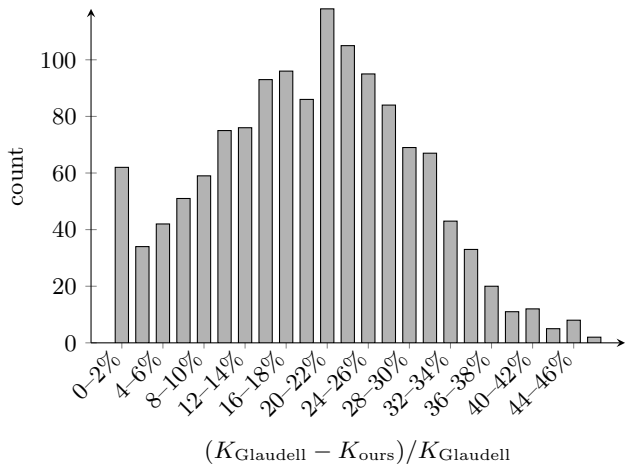

FIG. 3. Distribution of the relative K-count reduction $(K_{\text{Glaudell}} - K_{\text{ours}})/K_{\text{Glaudell}}$, binned in 2 % increments. Our method improves on Glaudell et al.'s K-count in 95.5 % of cases, with a mean reduction of 19.4 %, a median of 19.8 %, and a maximum of 47.7 %.

the distribution of the relative K-count reduction. From the data, we observe the following.

A. CS-count agreement

Across all 1,346 operators, our *CS*-count and Glaudell et al.'s *CS*-count are identical. So our algorithm is also *CS*-optimal in practice.

B. K-count improvement

Our algorithm produces a K-count less than or equal to that of Glaudell et al. in every case. It strictly improves the K-count in 95.5% of the cases; see Figure 3.

C. Tighter bound of the optimal CS-count

The inequalities $\text{lde} - 2 \leq CS\text{-count} \leq K_{\text{ours}}$ hold for every operator in the dataset. In particular,

- $\text{lde} \leq CS$-count: 1,324 cases (98.4%),
- $\text{lde} > CS$-count: 22 cases (1.6%).

In the 22 cases, the deficit is usually 1, with the largest gap being 2 for $\text{lde} = 20$, $CS\text{-count} = 18$.

D. Source of improvement

Two tight linear relationships emerge from the data. First, the K-count of Glaudell et al. is well approximated by twice the *CS*-count:

$$K_{\text{Glaudell}} \approx 2 \cdot CS\text{-count}.$$

The difference lies in $\{-4, -3, -2, -1, 0, 1, 2\}$, with 93.5% of values in $\{-4, \ldots, 0\}$. Second, our K-count is well approximated by twice the lde:

$$K_{\text{ours}} \approx 2 \cdot \text{lde}.$$

The difference lies in $\{-2, -1, 0\}$, with $\{-1, 0\}$ accounting for 80.5% of cases. These two relationships are also reflected in the Pearson correlations: the correlation between lde and K_{ours} is 0.9996, while the correlation between *CS*-count and K_{Glaudell} is 0.9991. Since $K_{\text{Glaudell}} \approx 2 \cdot CS\text{-count}$ and $K_{\text{ours}} \approx 2 \cdot \text{lde}$, the absolute K-count saving is approximately $2 \cdot (CS\text{-count} - \text{lde})$. In other words, the improvement is governed by the gap between the *CS*-count and the lde. The relative reduction percentage is roughly stable across lde values, hovering around 19–21% for lde between 5 and 40, and tapering off for very small or very large lde.

VI. CODE AND DATA AVAILABILITY

Code and data are available from [7].

VII. CONCLUSION AND FUTURE WORK

In conclusion, we gave an efficient and near-optimal algorithm for the exact synthesis of two-qubit Clifford+*CS* circuits from a given unitary matrix. In theory, the CS-count is at worst 4 times the optimal, but in practice, our method always achieves the optimal CS-count. In theory, the K-count is at worst 2 times the optimal, and in practice, our method always achieves a better K-count than the best known.

The algorithm of Li et al. [30] outputs a circuit whose T-count is at worst 10 times the optimal T-count. We believe our method can be used to improve this bound.

VIII. ACKNOWLEDGEMENTS

The authors acknowledge support from the National Natural Science Foundation of China under Grant 92465202. We have used Claude Opus 4.6 for

- creating four references [12, 15, 28, 29] and their one-line summary in the related work section.
- improving the wording of the paper.
- generating Haskell code: the ParseData.hs (for parsing data and drawing distribution diagrams) and CliffordCS.hs (the implementation of Glaudell et al.'s algorithm [17]).
- drafting Section VI, with proper guide from the authors (later revisions are made).

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